

4.6.4 Logic gates

4.6.4.1 Logic gates

Content	Additional information
Construct truth tables for the following logic gates: • NOT • AND • OR • XOR • NAND • NOR.	Students should know and be able to use ANSI/IEEE standard 91-1984 Distinctive shape logic gate symbols for these logic gates.
Be familiar with drawing and interpreting logic gate circuit diagrams involving one or more of the above gates.	
Complete a truth table for a given logic gate circuit.	
Write a Boolean expression for a given logic gate circuit.	
Draw an equivalent logic gate circuit for a given Boolean expression.	
Recognise and trace the logic of the circuits of a half-adder and a full-adder.	
Construct the circuit for a half-adder.	
Be familiar with the use of the edge-triggered D-type flip-flop as a memory unit.	Knowledge of internal operation of this flip-flop is not required.